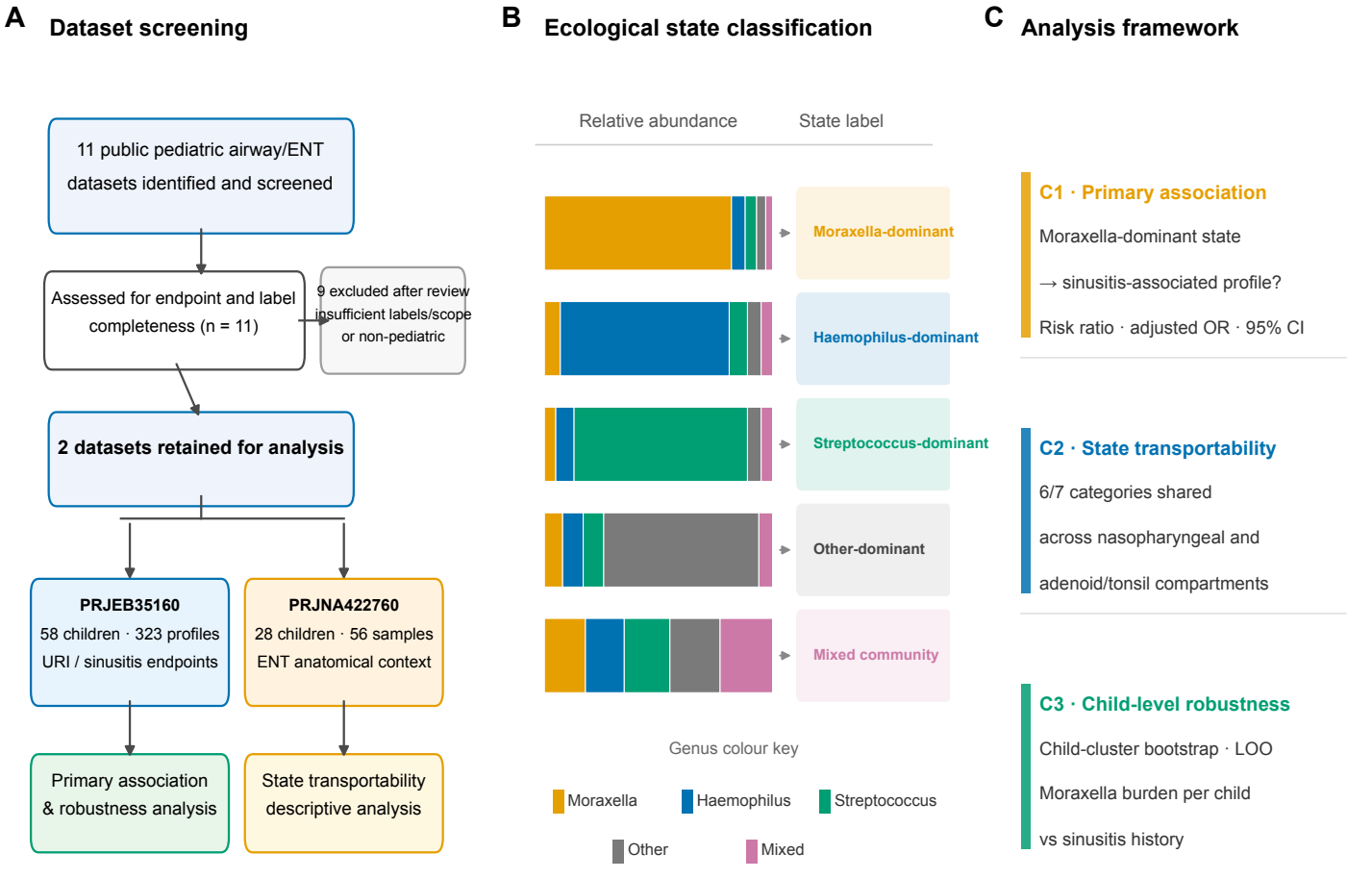
A
Dataset screening
11 public pediatric airway/ENT datasets identified and screened
Assessed for endpoint and label completeness (n = 11)
9 excluded after review insufficient labels/scope or non-pediatric
2 datasets retained for analysis
PRJEB35160
58 children · 323 profiles
URI / sinusitis endpoints
PRJNA422760
28 children · 56 samples
ENT anatomical context
Primary association & robustness analysis
State transportability descriptive analysis
B
Ecological state classification
Relative abundance
State label
Moraxella-dominant
Haemophilus-dominant
Streptococcus-dominant
Other-dominant
Mixed community
Genus colour key
Moraxella
Haemophilus
Streptococcus
Other
Mixed
C
Analysis framework
C1 · Primary association
Moraxella-dominant state
→ sinusitis-associated profile?
Risk ratio · adjusted OR · 95% CI
C2 · State transportability
6/7 categories shared
across nasopharyngeal and
adenoid/tonsil compartments
C3 · Child-level robustness
Child-cluster bootstrap · LOO
Moraxella burden per child
vs sinusitis history